Existing TA6586: paired outputs 5+6 and 7+8. Carrier is not a complete power/protection circuit.

FI/BI pulldowns are on the IC side of series resistors. Motor current never returns through MCU wiring.

VM/DRIVE_GND from a separately qualified supply path. No direct battery/charger ground equivalence is implied.

7 A is NOT a board rating. Copper, current, temperature, power sequencing and actual logic levels need bench checks.

COMBINED PROPOSAL: motor at left, logic at right. No physical power/thermal qualification or fabrication release.

H8 = qualified Waveshare +5V. H9 = XIAO BAT0 positive. H10 = XIAO ground. H5 = motor supply return. H3 = Waveshare ground.

VM remains the separately qualified motor supply. Never bridge VM, WAVE_5V, XIAO_BAT or raw battery terminals.

U2 TPS73701 and U3 MAX40200 are on the underside. Package/paste/via manufacture and actual PCB fit remain open.

H4 VM

H5 DRIVE_GND

H10 DRIVE_GND

U1 TA6586 DIP8

C1 100n / 25V

C2 470u / 16V

H6 FO

H7 BO

H1 CMD_FI

H2 CMD_BI

H3 DRIVE_GND

R1 100 / 1%

R3 10k / 1%

R2 100 / 1%

R4 10k / 1%

U2 TPS73701DRBR

U3 MAX40200AUK+T

H8 WAVE_5V

H9 XIAO_BAT

R5 23.2k / 0.1%

R6 10k / 0.1%

R7 360 / 1% / 0.25W

C3 4.7u / 10V X7R

C4 4.7u / 10V X7R

C5 1u / 10V X7R

C6 1u / 10V X7R

vea.raceremote

Sheets: /

File: drive-power-carrier.kicad_sch

Title: TA6586 + XIAO logic power — COMBINED PCB PROPOSAL

Size: A1 | Date: 2026-09-15 | **Rev: 0.3**

KiCad E.D.A. 10.0.6 | Id: 1/1